Machine Learning Classification Model

Auspify Internship — Task 5 Report

1. Objective

The objective of this task was to build and compare machine learning classification models for predicting whether a Netflix title is a Movie or TV Show.

2. Methodology

Relevant features were prepared from the Netflix dataset, missing values were handled, and the data was divided into training and testing sets using an 80/20 split.

Three classification algorithms were trained: Logistic Regression, Decision Tree, and Random Forest. Their performance was evaluated using Accuracy, Precision, Recall, and F1-Score.

3. Model Performance

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	99.83%	99.63%	99.81%	99.72%
Decision Tree	99.72%	99.62%	99.44%	99.53%
Random Forest	99.77%	99.62%	99.62%	99.62%

4. Findings

All three models achieved very high performance. Logistic Regression was the best-performing model, recording 99.83% accuracy and the highest F1-Score of 99.72%. Random Forest followed closely, while Decision Tree had the lowest overall performance.

5. Conclusion

The task successfully demonstrated the use of machine learning for Netflix content classification. Based on the evaluation results, Logistic Regression was the strongest of the tested models for classifying titles as Movies or TV Shows.

6. Skills Demonstrated

Python, Pandas, Scikit-learn, feature preparation, train-test splitting, classification algorithms, model evaluation, predictive analytics, and model comparison.

Task 5 Status: Complete